ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH

ELECTRICAL and ELECTRONICS ENGINEERING
4th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Course Structure
(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Electrical and Electronics Engineering
Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	EEE241401	PCC	Control Systems-I	3	0	0	3	30	70
2	EEE241402	PCC	Electric Machines-II	3	0	0	3	30	70
3	EEE241403	PCC	Microprocessors and Embedded Systems	3	0	0	3	30	70
4	EEE241404	PCC	Electromagnetic Waves and Radiation	3	0	0	3	30	70
5	EEE241405	PCC	Analog Electronics Circuits	3	0	0	3	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	EEE241411	PCC	Control Systems-I Lab	0	0	2	1	15	35
9	EEE241412	PCC	Electric Machines-II Lab	0	0	2	1	15	35
10	EEE241413	PCC	Microprocessors and Embedded Systems Lab	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				**21**	**0**	**10**	**23**	**340**	**560**
Total Contact Hours per week: 31									
Total Credit: 23									

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241401	**Control Systems-I**	**3-0-0**	**3**

Prerequisites: Laplace transforms techniques

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Define, classify and compare different types of control systems
- **CO2**: Derive transfer function of control systems
- **CO3**: Analyze and determine the time response of control systems
- **CO4**: Analyze stability of control systems using analytical and graphical techniques
- **CO5**: Apply analytical and graphical techniques to design control systems

MODULE 1: Concepts of Control Systems (4 Lectures)

Definition, open loop and closed loop systems, definitions and examples of linear, non-linear,time-invariant and time variant, continuous and discrete control system, block diagram representation of control systems.

MODULE 2: Models of Physical Systems (7 Lectures)

Transfer function: definition and properties, poles, zeros and pole-zero map, formulation of differential equations for physical systems and derivation of transfer function: mechanical and electrical systems, derivation of transfer function using block diagrams reduction techniques and signal flow graphs, signal flow graph from block diagram, analogous systems.

MODULE 3: Introduction to control system components (3 Lectures)

Controller components, potentiometer, AC/DC servomotors, tacho-generator, synchros, stepper motor.

MODULE 4: Time-domain analysis (7 Lectures)

Concept of transient response and steady-state response, standard test signals - step, ramp, parabolic and impulse signals, time response of first order and second order systems, closed loop transfer function, characteristic equation, performance specifications in time domain, derivative and integral control and their effects on the performance of the 2nd order systems, system types and error constants, generalized error coefficients, transient response of higher order systems

MODULE 5: Stability analysis (5 Lectures)

Concepts of control system stability, relation between stability and pole locations, Routh-Hurwitz stability criterion, scopes and limitations of the criterion, root-locus techniques, system analysis and design using root-locus technique

MODULE 6: Frequency response analysis (8 Lectures)

Frequency response and its specifications, stability analysis using frequency response plots: Bode plot, polar plot, log-magnitude vs phase plots, Nyquist plot and Nyquist stability criterion, M and N circle.

MODULE 7: Compensation Techniques (6 Lectures)

Preliminary design specifications in time and frequency domain, gain compensation, lead and lag compensation.

Text Books/References:

1. Nagrath, I. J.& Gopal, M, Control Systems Engineering, New Age International Publishers, 8th Edition, 2025
2. Ogata, K., Modern Control Engineering, Pearson Education, 5th Edition, 2015
3. Kuo, Benjamin C, Automatic Control Systems, Prentice Hall of India, 7th Edition, 2013
4. Golnaraghi, F.&Kuo, Benjamin C., Automatic Control Systems, McGraw Hill Education (India) Pvt. Ltd., 10th Edition, 2018

5. Rao V. Dukkipati, Analysis and Design of Control systems using MATLAB, New Age International Publishers, 2nd Edition
6. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241402	**Electrical Machines-II**	**3-0-0**	**3**

Prerequisites: Electrical Machine-I

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: To apply the knowledge of electro-magnetic to explain the working of an induction motor.
- **CO2**: To have the knowledge of working principles of alternators and their role in electrical power generation. Also, they will know about the characteristics of synchronous motors and their applications.
- **CO3**: To apply the knowledge of equivalent circuit, basic equations for determination of torque, power and efficiency and to perform laboratory test for determination of motor parameters.
- **CO4**: To have knowledge of principle of operation of various motors, such as universal motor, repulsion motor, reluctance motor, stepper motor and BLDC motor, and their field of applications.
- **CO5**: To analyze the performance of induction motors based on their control and applications.

MODULE 1: Fundamentals of AC Machine Windings (7 Lectures)

Physical arrangement of windings in stator and cylindrical rotor; slots for windings, single turn coil - active portion and overhang, full-pitch and short-pitch coils, concentrated winding, distributed winding, winding axis, 3D visualization of the above winding types, winding factors. Air-gap MMF distribution with fixed current through - concentrated and distributed and Sinusoidally distributed winding,

MODULE 2: Poly-phase Induction Machines (12 Lectures)

Construction, Types (squirrel cage and slip-ring). Three windings spatially shifted by 120 degrees (carrying three-phase balanced currents), revolving magnetic field. Operating principle of poly-phase induction motors. Torque Slip Characteristics, Starting and Maximum Torque, Equivalent circuit, Phasor Diagram, Losses and Efficiency, Effect of parameter variation on torque speed characteristics (variation of rotor and stator resistances, stator voltage, frequency), Methods of starting, braking and speed control for induction motors; Generator operation, Self-excitation, Doubly-fed Induction Machines.

MODULE 3: Single-phase Induction Motors (6 Lectures)

Constructional features of single phase induction motor. Magnetic field produced by a single winding - fixed current and alternating current. Double revolving field theory, equivalent circuit, determination of parameters, Split-phase starting methods and applications

MODULE 4: Synchronous Machines (15 Lectures)

Construction and principles of operation of synchronous generators, emf equation. Armature reaction, leakage reactance, synchronous reactance, and impendence of non-salient pole machines. Short circuit and open circuit tests, short circuit ratio, M M F in salient and non-salient pole machines. Calculation of regulation by synchronous impedance method. MMF method and ASA method.

Introduction to two-reactance theory, locus diagram of synchronous impedance, slip test, damper winding and oscillation of synchronous machines, Synchronization, power angle diagram and synchronizing power. Sub-transient and transient reactance of synchronous machine. Parallel operation and load sharing of synchronous machines. Construction and principles of operation of synchronous motor. Phasor diagram, effect of varying excitation, effect of load variation, V-curve, O-curve, power angle diagram and stability, Hunting, Two-reaction theory of salient-pole motor, Starting. Use as synchronous phase modifiers.

Text Books/References:

1. E. Fitzgerald and C. Kingsley, Electric Machinery, McGraw Hill Education, 2013
2. P. S. Bimbhra, Electrical Machinery, Khanna Publishers, 2011
3. M. N. Bandopadhyay, Electrical Machines: Theory and Practice, PHI
4. I.J. Nagrath and D. P. Kothari, Electric Machines, McGraw Hill Education, 2010
5. A. S. Langsdorf, Alternating current machines, McGraw Hill Education, 1984
6. B. L. Theraja, A. K. Theraja, A Text Book of Electrical Technology Vol II A.C. and D.C. Machines, S. Chand & Co., 2015
7. J. B. Gupta, Theory & Performance of Electrical Machines, S. K. Kataria & Sons, 2012
8. V.K. Mehta, Rohit Mehta, Principles of Electrical Machines, S. Chand & Co., 2009

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241403	**Microprocessors and Embedded Systems**	**3-0-0**	**3**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Explain the architecture of 8085 Microprocessor and some advanced processors.
- **CO2**: Apply assembly language for programming of 8085 Microprocessor.
- **CO3**: Describe the architecture of 8051 and ARM Microcontrollers.
- **CO4**: Make use of Assembly language and Embedded C for programming 8051 microcontroller.
- **CO5**: Design suitable interfacing of peripherals with 8085 microprocessor and 8051microcontroller.

MODULE 1: Architecture of Microprocessors (5 Lectures)

General processor architecture, 8085 architectures, Memory classification and interfacing, Memory – map, Address decoding (Absolute and partial); Pin diagram of 8085, interfacing of I/O devices, peripheral I/O and memory mapped I/O.

MODULE 2: Programming of 8085 (10 Lectures)

8085 instruction format and classification, addressing modes; Assembly language programming of 8085, hand assembly of a program. Program looping, counter, subroutine & linkages, delay routine. Instruction cycle, Machine cycle and T–state, Bus Timing diagram. Stack and subroutine, stack used by the microprocessor during CALL and RET instructions, stack used by the programmer, 8085 interrupt process, software and hardware interrupts, their priorities, concepts of Direct Memory Access.

MODULE 3: Interfacing Devices (7 Lectures)

Study of Programmable Interfacing devices like 8255, 8254, 8257, 8279 and their applications. Interfacing of LED and switches, matrix keyboard and multiplexed displays

Advanced Processor: Concepts of virtual memory, Cache memory, Advanced coprocessor Architectures- 286, 486, Pentium

MODULE 4: 8086 Architecture (7 Lectures)

Architecture, pin description, physical address, segmentation, memory organization, addressing modes, basic programming.

MODULE 5: Architecture of Microcontrollers (8 Lectures)

Overview of the architecture of 8051 microcontrollers, ARM microcontroller. RISC vs. CISC architecture. Addressing modes of 8051 and ARM 9 Microcontroller, Interrupt, Counter and Timer.

MODULE 6: Programming and Interfacing with 8051 and ARM microcontroller (3 Lectures)

Programming with Assembly and Embedded C. Interfacing with peripherals - timer, serial I/O, parallel I/O, A/D and D/A converters, 7 segment LEDs, LCDs, etc.

Text Books/References:

1. Ramesh Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram International Publishing Pvt. Ltd, 2013
2. Muhammad Ali Mazidi, Janice Gillispie Mazidi and Rolin D. McKinley, The 8051 Microcontroller and Embedded Systems using Assembly and C, PearsonEducation 2nd Edition, 2008
3. Doughlas.V.Hall, Microprocessor and Interfacing: Programming and Hardware, McGraw Hill, 1991
4. Manish K Patel, The 8051 microcontroller Based Embedded Systems, McGraw Hill, 2014
5. Sunil Mathur, Microprocessor 8086: Architecture, Programming and Interfacing, PHI, 2010
6. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241404	Electromagnetic Waves and Radiation	3-0-0	3

Prerequisites: To provide exposure to students to the principles governing Electromagnetics, working, radiating systems, waveguides, transmission lines and antenna and the respective applications.

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Students will be able to explain the fundamental concepts of Electromagnetic fields, Maxwell's equation, uniform plane waves in free space and other mediums.
- **CO2**: Students will be able to learn the working of transmission lines and waveguides.
- **CO3**: Students will be able to learn the working principle of waveguides.

MODULE 1: Vector Analysis (3 Lectures)
Review of dot and cross products, gradient, divergence and curl. Divergence and Stoke's Theorem, Cartesian, Cylindrical and Spherical Co-ordinate system. Transformation between coordinates, General curvilinear coordinates. Value of gradient, divergence and curl in general co-ordinates and to obtain their values in cylindrical and spherical coordinates.

MODULE 2: The Static Electric Field (9 Lectures)
Coulomb's law, Electric field strength, Field due to point charges, a line charge and a sheet of charge, Field due to continuous volume charge, Electric flux density, Gauss's law in integral form, Gauss's law in differential form (Maxwell's first equation in electrostatics), Applications of the Gauss's law. Electrostatic potential difference and potential, potential and potential difference expressed as a line integral potential field of a point charge, potential field of a system of charges, conservative property, potential gradient, the dipole, energy density in the electrostatic field.

MODULE 3: The Static Magnetic Field (9 Lectures)
The Biot-Savart's law (the magnetic field of filamentary currents), the magnetic field of distributed surface and volume currents, Ampere's Circuital law in integral and differential form(Maxwell's curl equation for steady magnetic field),The scalar and vector magnetic potentials, Maxwell's divergence equation for B, steady magnetic field laws, forces in magnetic field, force on a current element, force between two current elements, force and torque in the current loop.

MODULE 4: The Electromagnetic Field (7 Lectures)
Faraday's law in integral and differential form (Maxwell's first curl equation for electromagnetic field The Lorentz force equation. The concept of displacement current and modified Ampere's law (Maxwell's 2nd curl equation for electromagnetic field), The continuity equation, power flow in an electromagnetic field, Poynting Vector. Sinusoidally time varying fields, Maxwell's equations for sinusoidally time varying fields, power and energy considerations for sinusoidally time varying fields. The retarded potentials, Polarization of vector fields, review of Maxwell's equations.

MODULE 5: Materials and Fields (6 Lectures)
Current and current density, the continuity equation, conductor in fields. Dielectric in fields: Polarization, flux density, electric susceptibility, relative permittivity, Boundary conditions in perfect dielectrics, magnetic materials, magnetization, permeability, boundary conditions.

MODULE 6: Applied Electromagnetics (6 Lectures)
Poisson's and Laplace equations, solution of one-dimensional cases, general solution of Laplace equation, Method of images. Electromagnetic waves, The Helm Holtz equation, wave motion and free space, wave motion in perfect lossy dielectrics, propagation in good conductors, skin effect, Reflection of uniform plane waves. Radiation of electromagnetic waves.

Text Books/References:

1. N.N. Rao, Elements of Engineering Electromagnetics, Pearson Education, 2008
2. D.K. Cheng, Field and Wave Electromagnetics, Pearson Education,
3. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241405	**Analog Electronics Circuits**	**3-0-0**	**3**

Prerequisites: CIRCUITS of Engineering Systems
Course Outcomes: At the end of the course the students will be able to

- **CO1:** Design and analyze various amplifier circuits.
- **CO2:** Design sinusoidal and non-sinusoidal oscillators.
- **CO3:** Design OP-AMP based circuits.

MODULE 1: MOSFET Circuits (5 Lectures)
MOSFET structure and I-V characteristics. MOSFET as a switch. MOSFET as an amplifier: small-signal model and biasing circuits, common-source, common-gate and common-drain amplifiers; small signal equivalent circuits - gain, input and output impedances, trans- conductance, high frequency equivalent circuit.

MODULE 2: Amplifiers (12 Lectures)
Amplifier Models: Voltage amplifier, current amplifier, trans-conductance amplifier and trans-resistance amplifier.
Single Stage Amplifier: Small signal analysis, low frequency transistor models, estimation of voltage gain, input resistance, output resistance, design procedure for particular specifications, High frequency transistor models, high frequency response of single stage amplifier. Current mirror: Basic topology and its variants, V-I characteristics
Multistage Amplifiers: Classification, Distortion, Low and high Frequency responses and Bode plots, Cascade amplifiers (CE-CB), effect of coupling and bypass capacitors, RC coupled amplifier and its low frequency response.
Power Amplifiers: Various classes of operation (Class A, B, AB and C), their power efficiency and linearity issues.

MODULE 3: Feedback Topologies (7 Lectures)
Voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., calculation with practical circuits, concept of stability, gain margin and phase margin.

MODULE 4: Oscillators (7 Lectures)
Barkhausen criterion, RC oscillators (phase shift, Wien bridge), LC oscillators (Hartley, Colpitt, Clapp), crystal oscillator.

MODULE 5: OP-AMP Applications (6 Lectures)
Linear and Non-linear applications: Review of inverting and non-inverting amplifiers, integrator and differentiator, summing amplifier, difference amplifier, instrumentation amplifier, Voltage to current converter, current to Voltage converter, Comparator, Window detectors, Sample and Hold Circuits, Peak Detectors, Precision rectifier, Log and antilog amplifiers, Schmitt trigger and its applications.
Active Filters: Low pass, high pass, band pass and band stop.

MODULE 6: Differential Amplifier (3 Lectures)
Basic structure and principle of operation, calculation of differential gain, common mode gain, CMRR and ICMR. OP-AMP design: Ideal OPAMP characteristics and limitations, Design of differential amplifier for a given specification, design of gain stages and output stages, compensation.

Text Books/References:

1. "Introduction to Operational Amplifier theory and applications", J.V. Wait, L.P. Huelsman and GA Korn, McGraw Hill, 1992.

2. Electronic Devices and Circuit Theory, R. L. Boylestad and L. Nashelsky, 11th edition. Pearson, 2013.
3. Electronic Circuits: Analysis and Design, D. A. Neamen, 3rd edition. Tata McGraw-Hill, 2008

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	**Finance and Accounting**	**3-0-0**	**3**

Prerequisites: High School Education
Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash
- books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting
Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation
Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank
Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts
Concept of Capital Expenditure and Revenue Expenditure, Bad Debts,
Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment
Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market-Initial Public Offering , Market Players, Secondary Market :Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns-(Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	**Environmental Science**	**3-0-0**	**0**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Relate environment and its associated problems
- **CO2:** Explain basic ethics of living and protection of other organisms
- **CO3:** Applying sustainable development principles

MODULE 1: Introduction
Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse
Land use: Impact of land – use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution
Introduction: Water quality standards, Water pollution

MODULE 4: Air Pollution
Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5: Noise Pollution
Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6: Control of Pollutions
Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241411	**Control Systems-I Lab**	**0-0-2**	**1**

Prerequisites: Linear Control Systems
Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Solve control system problems using MATLAB software
- **CO2**: Analyze the response of control system by measuring relevant parameters
- **CO3**: Interpret the role of various components in control system
- **CO4**: Compare theoretical predictions with experimental results to resolve any apparent differences

Module No.	Description	Contact Hours
PART I:	Use of MATLAB software for solving Control System Problems	10
PART II:	To perform experiments on control systems using experimental kits **LIST OF EXPERIMENTS:** 1. Light Intensity Control Systems 2. DC Position Control Systems 3. Potentiometer Error Detector 4. Speed-Torque Characteristics of AC Servomotor 5. Synchro-Transmitter Control Transformer pair as an Error Detector	10
	Total Contact Hours	20

Text Books/References:

1. Mastering MATLAB, Pearson Education
2. Rao V. Dukkipati, Analysis and Design of control systems using MATLAB, New Age International Publishers, 2nd Edition

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241412	**Electrical Machines-II Lab**	**0-0-2**	**1**

Prerequisites: Knowledge of electrical circuits, electromagnetic induction and magnetic circuits
Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Explain the mode of starting and switching-off with precautionary measures while handling different types of electrical machines
- **CO2**: Apply the theoretical concepts for experimentation with different electrical machines
- **CO3**: Develop report writing skills

LIST OF EXPERIMENTS

Module No.	Description	Contact Hours
1	Retardation test on a D.C. Machine	3
2	No-load test and blocked-rotor test on 3-phase induction motor	3
3	Working of Single phase and three-phase Induction Regulator	3
4	V-Curves of a synchronous motor	3
5	Slip-Test on Alternator	3
6	Synchronization of alternators	3
	Total Contact Hours	18

Course Code	Course Title	Hours per week L-T-P	Credit
EEE241413	**Microprocessors and Embedded Systems Lab**	**0-0-2**	**1**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Understand the architecture and the instruction set of 8085 microprocessors
- **CO2**: Design and develop programs using 8085 microprocessor kit and 8051 microcontrollers
- **CO3**: Write an engineering report in an organized way

LIST OF EXPERIMENTS

Experiment	Description	Contact Hours
1	**Familiarization of the Microprocessor Kit:** a) editing a program (b) Verifying the program (c) Executing the program and verifying the outcome of the program.	2
2	**Programming of 8085:** Developing and testing simple program for data transfer, block transfer, sorting of data, arithmetic operations, interrupt,	6
3	Interfacing 8255, traffic light control system	4
4	**Programming of 8051 using embedded C:** Develop simple programs for implementation in 8051 systems	8
	Total Contact Hours	20

Text Books/References:

1. Ramesh Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram International Publishing Pvt. Ltd, 2013

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	**Micro Project (Skill Based)**	**0-0-4**	**2**

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- CO1: Analyze engineering problems and their solutions
- CO2: Apply modern engineering tools to attempt solutions to the problems
- CO3: Adapt with computational thinking and automation skills
- CO4: Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5

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